

| BRK (00) |                                      | Cycles: 7                                                                                                   | Size: 2 * |
|----------|--------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------|
| Implied  |                                      |                                                                                                             |           |
| Cycle    | R/W                                  | Desc.                                                                                                       |           |
| -X.2     | Read <b>PC:PB</b>                    | ** Store as OpCode.                                                                                         |           |
| 1.1      |                                      | Inc. <b>PC</b> .                                                                                            |           |
| 1.2      | Read <b>PC:PB</b>                    | Store as Operand.                                                                                           |           |
| 2.1      |                                      | Inc. <b>PC</b> .                                                                                            |           |
| 2.2      | *** Write to ( <b>S</b> +0)   \$0100 | Push <b>PC</b> .H onto stack.                                                                               |           |
| 3.1      |                                      |                                                                                                             |           |
| 3.2      | *** Write to ( <b>S</b> -1)   \$0100 | Push <b>PC</b> .L onto stack.                                                                               |           |
| 4.1      |                                      | Use Interrupt flags to decide which vector to use.<br>RESET = \$FFFC.<br>NMI = \$FFFA.<br>IRQ/BRK = \$FFFE. |           |
| 4.2      | *** Write to ( <b>S</b> -2)   \$0100 | Push <b>P</b> onto stack with <b>B</b> flag if software BRK.                                                |           |
| 5.1      |                                      | Dec. <b>S</b> by 3.                                                                                         |           |
| 5.2      | Read Vector                          | Store to Address.L.                                                                                         |           |
| 6.1      |                                      | Set <b>I</b> , unset <b>D</b> .                                                                             |           |
| 6.2      | Read Vector + 1                      | Store to Address.H.                                                                                         |           |
| 7.1      |                                      | Enable writes to PC (disabled by NMI/IRQ). Copy Address to PC. Set PB to 0.                                 |           |
| 7.2      | Read <b>PC:PB</b>                    | Store as OpCode.                                                                                            |           |
| +X.1     |                                      |                                                                                                             |           |

\* Concerning the BRK instruction, you should also note that although its second byte is basically a “don’t care” byte – that is, it can have any value - the BRK (and COP instruction as well) is a two-byte instruction, the second byte sometimes is used as a signature byte to determine the nature of the BRK being executed. When an RTI instruction is executed, control always returns to the second byte past the BRK opcode. — assembly-programming-manual-for-w65c816.pdf

\*\* If NMI, IRQ, or RESET, OpCode is forced to 0 and writes to PC are disabled.

\*\*\* If handling a RESET, the writes turn into reads. The databus is populated but ignored.

| ORA (01) Cycles: 6                   |                   | Size: 2                                                                                                         |
|--------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Read) |                   |                                                                                                                 |
| Cycle                                | R/W               | Desc.                                                                                                           |
| -X.2                                 | Read <b>PC:PB</b> | Store as OpCode.                                                                                                |
| 1.1                                  | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                |
| 1.2                                  |                   | Store as Pointer.                                                                                               |
| 2.1                                  |                   | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> . Address.L = Pointer.L. Address.H = <b>D</b> .H. |
| 2.2                                  |                   | If <b>D</b> .L is 0, skip the next cycle.                                                                       |
| 3.1                                  |                   | Address.H = Pointer.H (fixes high byte of address).                                                             |
| 3.2                                  |                   |                                                                                                                 |
| 4.1                                  | Read Address      | Store as Pointer.                                                                                               |
| 4.2                                  |                   |                                                                                                                 |
| 5.1                                  | Read Address + 1  | Store as Pointer.H.                                                                                             |
| 5.2                                  |                   |                                                                                                                 |
| 6.1                                  | Read Pointer:DB   | Store as Operand.                                                                                               |
| 6.2                                  |                   |                                                                                                                 |
| 7.1                                  |                   |                                                                                                                 |
| 7.2                                  | Read <b>PC:PB</b> | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L.   |
|                                      |                   | Store as OpCode.                                                                                                |
| +X.1                                 |                   |                                                                                                                 |

|          |                                  |                                                 |
|----------|----------------------------------|-------------------------------------------------|
| COP (02) | Cycles: 7                        | Size: 2                                         |
| Implied  |                                  |                                                 |
| Cycle    | R/W                              | Desc.                                           |
| -X.2     | Read <b>PC:PB</b>                | Store as OpCode.                                |
| 1.1      |                                  | Inc. <b>PC</b> .                                |
| 1.2      | Read <b>PC:PB</b>                | Store as Operand.                               |
| 2.1      |                                  | Inc. <b>PC</b> .                                |
| 2.2      | Write to ( <b>S</b> +0)   \$0100 | Push <b>PC</b> .H onto stack.                   |
| 3.1      |                                  |                                                 |
| 3.2      | Write to ( <b>S</b> -1)   \$0100 | Push <b>PC</b> .L onto stack.                   |
| 4.1      |                                  | Set Vector to \$FFF4.                           |
| 4.2      | Write to ( <b>S</b> -2)   \$0100 | Push <b>P</b> onto stack.                       |
| 5.1      |                                  | Dec. <b>S</b> by 3.                             |
| 5.2      | Read Vector                      | Store to Address.L.                             |
| 6.1      |                                  | Set <b>I</b> , unset <b>D</b> .                 |
| 6.2      | Read Vector + 1                  | Store to Address.H.                             |
| 7.1      |                                  | Copy Address to <b>PC</b> . Set <b>PB</b> to 0. |
| 7.2      | Read <b>PC:PB</b>                | Store as OpCode.                                |
| +X.1     |                                  |                                                 |

|                       |                   |                                                                                                              |
|-----------------------|-------------------|--------------------------------------------------------------------------------------------------------------|
| ORA (03)              | Cycles: 4         | Size: 2                                                                                                      |
| Stack Relative (Read) |                   |                                                                                                              |
| Cycle                 | R/W               | Desc.                                                                                                        |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |
| 1.1                   | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                             |
| 1.2                   |                   | Store as Pointer.                                                                                            |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S.L</b>   \$100.                                                |
| 2.2                   |                   |                                                                                                              |
| 3.1                   |                   |                                                                                                              |
| 3.2                   | Read Address      | Store as Operand.                                                                                            |
| 4.1                   | Read <b>PC:PB</b> | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |
| 4.2                   |                   | Store as OpCode.                                                                                             |
| +X.1                  |                   |                                                                                                              |

|                            |                   |                                                                                   |
|----------------------------|-------------------|-----------------------------------------------------------------------------------|
| TSB (04)                   | Cycles: 5         | Size: 2                                                                           |
| Direct (Read/Modify/Write) |                   |                                                                                   |
| Cycle                      | R/W               | Desc.                                                                             |
| -X.2                       | Read <b>PC:PB</b> | Store as OpCode.                                                                  |
| 1.1                        |                   | Inc. <b>PC</b> .                                                                  |
| 1.2                        | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1                        |                   | Inc. <b>PC</b> .                                                                  |
| 2.2                        |                   |                                                                                   |
| 3.1                        |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2                        | Read Address      | Store as Operand.                                                                 |
| 4.1                        |                   |                                                                                   |
| 4.2                        | Write to Address  | Write Operand.L.                                                                  |
| 5.1                        |                   | Set <b>Z</b> based off ( <b>A.L</b> & Operand.L). Perform Operand  = <b>A</b> .   |
| 5.2                        | Write to Address  | Write Operand.L.                                                                  |
| 6.1                        |                   |                                                                                   |
| 6.2                        | Read <b>PC:PB</b> | Store as OpCode.                                                                  |
| +X.1                       |                   |                                                                                   |

|               |                   |                                                                                                              |         |
|---------------|-------------------|--------------------------------------------------------------------------------------------------------------|---------|
| ORA (05)      |                   | Cycles: 3                                                                                                    | Size: 2 |
| Direct (Read) |                   |                                                                                                              |         |
| Cycle         | R/W               | Desc.                                                                                                        |         |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |         |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                             |         |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                   |         |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                             |         |
| 2.2           |                   |                                                                                                              |         |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                            |         |
| 3.2           | Read Address      | Store as Operand.                                                                                            |         |
| 4.1           |                   | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |         |
| 4.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |         |
| +X.1          |                   |                                                                                                              |         |

|                            |                   |                                                                                                                                                 |
|----------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ASL (06)                   | Cycles: 5         | Size: 2                                                                                                                                         |
| Direct (Read/Modify/Write) |                   |                                                                                                                                                 |
| Cycle                      | R/W               | Desc.                                                                                                                                           |
| -X.2                       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                |
| 1.1                        |                   | Inc. <b>PC</b> .                                                                                                                                |
| 1.2                        | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                      |
| 2.1                        |                   | Inc. <b>PC</b> .                                                                                                                                |
| 2.2                        |                   |                                                                                                                                                 |
| 3.1                        |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                               |
| 3.2                        | Read Address      | Store as Operand.                                                                                                                               |
| 4.1                        |                   |                                                                                                                                                 |
| 4.2                        | Write to Address  | Write Operand.L.                                                                                                                                |
| 5.1                        |                   | Set <b>C</b> based off (Operand.L & \$80). Perform Operand.L <=< 1. Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L. |
| 5.2                        | Write to Address  | Write Operand.L.                                                                                                                                |
| 6.1                        |                   |                                                                                                                                                 |
| 6.2                        | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                |
| +X.1                       |                   |                                                                                                                                                 |

|                             |                   |                                                                                                              |
|-----------------------------|-------------------|--------------------------------------------------------------------------------------------------------------|
| ORA (07)                    | Cycles: 6         | Size: 2                                                                                                      |
| Direct Indirect Long (Read) |                   |                                                                                                              |
| Cycle                       | R/W               | Desc.                                                                                                        |
| -X.2                        | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |
| 1.1                         |                   | Inc. <b>PC</b> .                                                                                             |
| 1.2                         | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                   |
| 2.1                         |                   | Inc. <b>PC</b> .                                                                                             |
| 2.2                         |                   |                                                                                                              |
| 3.1                         |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                            |
| 3.2                         | Read Address      | Store as Pointer.                                                                                            |
| 4.1                         |                   |                                                                                                              |
| 4.2                         | Read Address + 1  | Store as Pointer.H.                                                                                          |
| 5.1                         |                   |                                                                                                              |
| 5.2                         | Read Address + 2  | Store as Bank.                                                                                               |
| 6.1                         |                   |                                                                                                              |
| 6.2                         | Read Pointer:Bank | Store as Operand.                                                                                            |
| 7.1                         |                   | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |
| 7.2                         | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |
| +X.1                        |                   |                                                                                                              |



|          |                                  |                                                                  |
|----------|----------------------------------|------------------------------------------------------------------|
| PHP (08) | Cycles: 3                        | Size: 1                                                          |
| Implied  |                                  |                                                                  |
| Cycle    | R/W                              | Desc.                                                            |
| -X.2     | Read <b>PC:PB</b>                | Store as OpCode.                                                 |
| 1.1      | Write to ( <b>S</b> +0)   \$0100 | Inc. <b>PC</b> .                                                 |
| 1.2      |                                  |                                                                  |
| 2.1      |                                  | Sets Operand.L to <b>P</b> with <b>X</b> and <b>M</b> flags set. |
| 2.2      |                                  | Push Operand.L onto stack.                                       |
| 3.1      |                                  | Dec. <b>S</b> by 1.                                              |
| 3.2      |                                  | Store as OpCode.                                                 |
| +X.1     |                                  |                                                                  |

|           |                   |                                                                                                                               |         |
|-----------|-------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (09)  |                   | Cycles: 2                                                                                                                     | Size: 2 |
| Immediate |                   |                                                                                                                               |         |
| Cycle     | R/W               | Desc.                                                                                                                         |         |
| -X.2      | Read PC:PB        | Store as OpCode.                                                                                                              |         |
| 1.1       | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                              |         |
| 1.2       |                   | Store as Operand.                                                                                                             |         |
| 2.1       |                   | Inc. <b>PC</b> . Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |         |
| 2.2       |                   | Store as OpCode.                                                                                                              |         |
| +X.1      |                   |                                                                                                                               |         |

|          |                   |                                                                                                                                                          |
|----------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASL (0A) | Cycles: 2         | Size: 1                                                                                                                                                  |
| Implied  |                   |                                                                                                                                                          |
| Cycle    | R/W               | Desc.                                                                                                                                                    |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                         |
| 1.1      |                   | Inc. <b>PC</b> .                                                                                                                                         |
| 1.2      | Read <b>PC:PB</b> | Discard.                                                                                                                                                 |
| 2.1      |                   | Set <b>C</b> based off ( <b>A</b> .L & \$80). Perform <b>A</b> .L <= 1. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                         |
| +X.1     |                   |                                                                                                                                                          |

|          |                                                                           |                              |
|----------|---------------------------------------------------------------------------|------------------------------|
| PHD (0B) | Cycles: 4                                                                 | Size: 1                      |
| Implied  |                                                                           |                              |
| Cycle    | R/W                                                                       | Desc.                        |
| -X.2     | Read <b>PC:PB</b>                                                         | Store as OpCode.             |
| 1.1      | Write to ( <b>S</b> +0)   \$0100<br><br>Write to (( <b>S</b>   \$0100)-1) | Inc. <b>PC</b> .             |
| 1.2      |                                                                           |                              |
| 2.1      |                                                                           |                              |
| 2.2      |                                                                           | Push <b>D</b> .H onto stack. |
| 3.1      |                                                                           |                              |
| 3.2      |                                                                           | Push <b>D</b> .L onto stack. |
| 4.1      |                                                                           |                              |
| 4.2      |                                                                           | Store as OpCode.             |
| +X.1     |                                                                           |                              |

|                              |                             |                                                                                  |
|------------------------------|-----------------------------|----------------------------------------------------------------------------------|
| TSB (0C)                     | Cycles: 6                   | Size: 3                                                                          |
| Absolute (Read/Modify/Write) |                             |                                                                                  |
| Cycle                        | R/W                         | Desc.                                                                            |
| -X.2                         | Read <b>PC:PB</b>           | Store as OpCode.                                                                 |
| 1.1                          |                             | Inc. <b>PC</b> .                                                                 |
| 1.2                          | Read <b>PC:PB</b>           | Store as Address.                                                                |
| 2.1                          |                             | Inc. <b>PC</b> .                                                                 |
| 2.2                          | Read <b>PC:PB</b>           | Store as Address.L.                                                              |
| 3.1                          |                             | Inc. <b>PC</b> .                                                                 |
| 3.2                          | Read Address: <b>DB</b>     | Store as Operand.                                                                |
| 4.1                          |                             |                                                                                  |
| 4.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                 |
| 5.1                          |                             | Set <b>Z</b> based off ( <b>A</b> .L & Operand.L). Perform Operand  = <b>A</b> . |
| 5.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                 |
| 6.1                          |                             |                                                                                  |
| 6.2                          | Read <b>PC:PB</b>           | Store as OpCode.                                                                 |
| +X.1                         |                             |                                                                                  |

|                 |                         |                                                                                                               |
|-----------------|-------------------------|---------------------------------------------------------------------------------------------------------------|
| ORA (0D)        | Cycles: 4               | Size: 3                                                                                                       |
| Absolute (Read) |                         |                                                                                                               |
| Cycle           | R/W                     | Desc.                                                                                                         |
| -X.2            | Read <b>PC:PB</b>       | Store as OpCode.                                                                                              |
| 1.1             |                         | Inc. <b>PC</b> .                                                                                              |
| 1.2             | Read <b>PC:PB</b>       | Store as Address.                                                                                             |
| 2.1             |                         | Inc. <b>PC</b> .                                                                                              |
| 2.2             | Read <b>PC:PB</b>       | Store as Address.L.                                                                                           |
| 3.1             |                         | Inc. <b>PC</b> .                                                                                              |
| 3.2             | Read Address: <b>DB</b> | Store as Operand.                                                                                             |
| 4.1             |                         | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |
| 4.2             | Read <b>PC:PB</b>       | Store as OpCode.                                                                                              |
| +X.1            |                         |                                                                                                               |

|                              |                             |                                                                                                                                                 |
|------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ASL (0E)                     | Cycles: 6                   | Size: 3                                                                                                                                         |
| Absolute (Read/Modify/Write) |                             |                                                                                                                                                 |
| Cycle                        | R/W                         | Desc.                                                                                                                                           |
| -X.2                         | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                |
| 1.1                          |                             | Inc. <b>PC</b> .                                                                                                                                |
| 1.2                          | Read <b>PC:PB</b>           | Store as Address.                                                                                                                               |
| 2.1                          |                             | Inc. <b>PC</b> .                                                                                                                                |
| 2.2                          | Read <b>PC:PB</b>           | Store as Address.L.                                                                                                                             |
| 3.1                          |                             | Inc. <b>PC</b> .                                                                                                                                |
| 3.2                          | Read Address: <b>DB</b>     | Store as Operand.                                                                                                                               |
| 4.1                          |                             |                                                                                                                                                 |
| 4.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                                                                                |
| 5.1                          |                             | Set <b>C</b> based off (Operand.L & \$80). Perform Operand.L <=< 1. Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L. |
| 5.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                                                                                |
| 6.1                          |                             |                                                                                                                                                 |
| 6.2                          | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                |
| +X.1                         |                             |                                                                                                                                                 |

|                      |                   |                                                                                                               |         |
|----------------------|-------------------|---------------------------------------------------------------------------------------------------------------|---------|
| ORA (0F)             |                   | Cycles: 5                                                                                                     | Size: 4 |
| Absolute Long (Read) |                   |                                                                                                               |         |
| Cycle                | R/W               | Desc.                                                                                                         |         |
| -X.2                 | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| 1.1                  |                   | Inc. <b>PC</b> .                                                                                              |         |
| 1.2                  | Read <b>PC:PB</b> | Store as Address.                                                                                             |         |
| 2.1                  |                   | Inc. <b>PC</b> .                                                                                              |         |
| 2.2                  | Read <b>PC:PB</b> | Store as Address.L.                                                                                           |         |
| 3.1                  |                   | Inc. <b>PC</b> .                                                                                              |         |
| 3.2                  | Read <b>PC:PB</b> | Stores as Bank.                                                                                               |         |
| 4.1                  |                   | Inc. <b>PC</b> .                                                                                              |         |
| 4.2                  | Read Address:Bank | Store as Operand.                                                                                             |         |
| 5.1                  |                   | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |         |
| 5.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| +X.1                 |                   |                                                                                                               |         |



|          |                   |                                                                                                                                                                     |
|----------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BPL (10) | Cycles: 2         | Size: 2                                                                                                                                                             |
| Relative |                   |                                                                                                                                                                     |
| Cycle    | R/W               | Desc.                                                                                                                                                               |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                    |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> . Decide to branch if <b>N</b> is 0.                                                                                                                 |
| 1.2      |                   | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> , BoundaryCrossed = Address.H != <b>PC</b> .H. If not branching or not BoundaryCrossed, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 4.2).                                                                                                    |
| 2.2      |                   |                                                                                                                                                                     |
| 3.1      |                   | Set <b>PC</b> .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).                                                                                |
| 3.2      |                   | Poll interrupts.                                                                                                                                                    |
| 4.1      |                   | Set <b>PC</b> .H to Address.H                                                                                                                                       |
| 4.2      |                   | Store as OpCode.                                                                                                                                                    |
| +X.1     |                   |                                                                                                                                                                     |

|                                      |                   |                                                                                                                                                                       |         |
|--------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (11)                             |                   | Cycles: 5                                                                                                                                                             | Size: 2 |
| Direct Indirect Indexed (d),y (Read) |                   |                                                                                                                                                                       |         |
| Cycle                                | R/W               | Desc.                                                                                                                                                                 |         |
| -X.2                                 | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                      |         |
| 1.1                                  |                   | Inc. <b>PC</b> .                                                                                                                                                      |         |
| 1.2                                  | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                            |         |
| 2.1                                  |                   | Inc. <b>PC</b> .                                                                                                                                                      |         |
| 2.2                                  |                   |                                                                                                                                                                       |         |
| 3.1                                  |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                     |         |
| 3.2                                  | Read Address      | Store as Pointer.                                                                                                                                                     |         |
| 4.1                                  |                   |                                                                                                                                                                       |         |
| 4.2                                  | Read Address + 1  | Store as Pointer.H.                                                                                                                                                   |         |
| 5.1                                  |                   | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16). |         |
| 5.2                                  |                   |                                                                                                                                                                       |         |
| 6.1                                  |                   |                                                                                                                                                                       |         |
| 6.2                                  | Read Pointer:Bank | Store as Operand.                                                                                                                                                     |         |
| 7.1                                  |                   | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> .                                                          |         |
| 7.2                                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                      |         |
| +X.1                                 |                   |                                                                                                                                                                       |         |

|                        |                         |                                                                                                       |         |
|------------------------|-------------------------|-------------------------------------------------------------------------------------------------------|---------|
| ORA (12)               |                         | Cycles: 5                                                                                             | Size: 2 |
| Direct Indirect (Read) |                         |                                                                                                       |         |
| Cycle                  | R/W                     | Desc.                                                                                                 |         |
| -X.2                   | Read <b>PC:PB</b>       | Store as OpCode.                                                                                      |         |
| 1.1                    |                         | Inc. <b>PC</b> .                                                                                      |         |
| 1.2                    | Read <b>PC:PB</b>       | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                            |         |
| 2.1                    |                         | Inc. <b>PC</b> .                                                                                      |         |
| 2.2                    |                         |                                                                                                       |         |
| 3.1                    |                         | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                     |         |
| 3.2                    | Read Address            | Store as Pointer.                                                                                     |         |
| 4.1                    |                         |                                                                                                       |         |
| 4.2                    | Read Address + 1        | Store as Pointer.H.                                                                                   |         |
| 5.1                    |                         |                                                                                                       |         |
| 5.2                    | Read Pointer: <b>DB</b> | Store as Operand.                                                                                     |         |
| 6.1                    |                         | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and Z based off <b>A.L</b> . |         |
| 6.2                    | Read <b>PC:PB</b>       | Store as OpCode.                                                                                      |         |
| +X.1                   |                         |                                                                                                       |         |

|                                        |                     |                                                                                                              |         |
|----------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------|---------|
| ORA (13)                               |                     | Cycles: 7                                                                                                    | Size: 2 |
| Stack Relative Indirect Indexed (Read) |                     |                                                                                                              |         |
| Cycle                                  | R/W                 | Desc.                                                                                                        |         |
| -X.2                                   | Read <b>PC:PB</b>   | Store as OpCode.                                                                                             |         |
| 1.1                                    | Read <b>PC:PB</b> . | Inc. <b>PC</b> .                                                                                             |         |
| 1.2                                    |                     | Store as Pointer.                                                                                            |         |
| 2.1                                    |                     | Inc. <b>PC</b> . Set Address to Pointer + <b>S.L</b>   \$100.                                                |         |
| 2.2                                    |                     |                                                                                                              |         |
| 3.1                                    |                     |                                                                                                              |         |
| 3.2                                    | Read Address        | Store as Pointer.                                                                                            |         |
| 4.1                                    | Read Address + 1    |                                                                                                              |         |
| 4.2                                    |                     | Store as Pointer.H.                                                                                          |         |
| 5.1                                    |                     | Perform Tmp = ui32(Pointer + <b>Y</b> ).                                                                     |         |
|                                        |                     | Pointer = ui16(Tmp).                                                                                         |         |
|                                        |                     | Bank = <b>DB</b> + (Tmp >> 16).                                                                              |         |
| 5.2                                    | Read Pointer:Bank   |                                                                                                              |         |
| 6.1                                    |                     |                                                                                                              |         |
| 6.2                                    |                     | Store as Operand.                                                                                            |         |
| 7.1                                    |                     | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |         |
| 7.2                                    |                     | Store as OpCode.                                                                                             |         |
| +X.1                                   |                     |                                                                                                              |         |

|                            |                   |                                                                                   |
|----------------------------|-------------------|-----------------------------------------------------------------------------------|
| TRB (14)                   | Cycles: 5         | Size: 2                                                                           |
| Direct (Read/Modify/Write) |                   |                                                                                   |
| Cycle                      | R/W               | Desc.                                                                             |
| -X.2                       | Read <b>PC:PB</b> | Store as OpCode.                                                                  |
| 1.1                        |                   | Inc. <b>PC</b> .                                                                  |
| 1.2                        | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1                        |                   | Inc. <b>PC</b> .                                                                  |
| 2.2                        |                   |                                                                                   |
| 3.1                        |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2                        | Read Address      | Store as Operand.                                                                 |
| 4.1                        |                   |                                                                                   |
| 4.2                        | Write to Address  | Write Operand.L.                                                                  |
| 5.1                        |                   | Set <b>Z</b> based off ( <b>A.L</b> & Operand.L). Perform Operand &= ~ <b>A</b> . |
| 5.2                        | Write to Address  | Write Operand.L.                                                                  |
| 6.1                        |                   |                                                                                   |
| 6.2                        | Read <b>PC:PB</b> | Store as OpCode.                                                                  |
| +X.1                       |                   |                                                                                   |

|                  |                   |                                                                                                                                                                                          |
|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA (15)         | Cycles: 4         | Size: 2                                                                                                                                                                                  |
| Direct, X (Read) |                   |                                                                                                                                                                                          |
| Cycle            | R/W               | Desc.                                                                                                                                                                                    |
| -X.2             | Read PC:PB        | Store as OpCode.                                                                                                                                                                         |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                         |
| 1.2              |                   | Store as Operand.                                                                                                                                                                        |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                         |
| 2.2              |                   |                                                                                                                                                                                          |
| 3.1              |                   | If <b>D.L</b> is 0, Address.L = Operand + <b>X.L</b> + <b>D.L</b> , Address.H = <b>D.H</b> , otherwise Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2              |                   |                                                                                                                                                                                          |
| 4.1              |                   |                                                                                                                                                                                          |
| 4.2              | Read Address      | Store as Operand.                                                                                                                                                                        |
| 5.1              | Read <b>PC:PB</b> | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> .                                                                             |
| 5.2              |                   | Store as OpCode.                                                                                                                                                                         |
| +X.1             |                   |                                                                                                                                                                                          |

| ASL (16)                      | Cycles: 6         | Size: 2                                                                                                                                                                                  |
|-------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read/Modify/Write) |                   |                                                                                                                                                                                          |
| Cycle                         | R/W               | Desc.                                                                                                                                                                                    |
| -X.2                          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                         |
| 1.1                           | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                         |
| 1.2                           |                   | Store as Operand.                                                                                                                                                                        |
| 2.1                           |                   | Inc. <b>PC</b> .                                                                                                                                                                         |
| 2.2                           |                   |                                                                                                                                                                                          |
| 3.1                           |                   | If <b>D.L</b> is 0, Address.L = Operand + <b>X.L</b> + <b>D.L</b> , Address.H = <b>D.H</b> , otherwise Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2                           |                   |                                                                                                                                                                                          |
| 4.1                           |                   |                                                                                                                                                                                          |
| 4.2                           | Read Address      | Store as Operand.                                                                                                                                                                        |
| 5.1                           | Write to Address  | Write Operand.L.                                                                                                                                                                         |
| 5.2                           |                   | Set <b>C</b> based off (Operand.L & \$80). Perform Operand.L <=< 1. Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L.                                          |
| 6.1                           |                   |                                                                                                                                                                                          |
| 6.2                           | Write to Address  | Write Operand.L.                                                                                                                                                                         |
| 7.1                           | Read <b>PC:PB</b> |                                                                                                                                                                                          |
| 7.2                           |                   | Store as OpCode.                                                                                                                                                                         |
| +X.1                          |                   |                                                                                                                                                                                          |

|                                           |                                                                                                    |                                                                                                              |         |
|-------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------|
| ORA (17)                                  |                                                                                                    | Cycles: 6                                                                                                    | Size: 2 |
| Direct Indirect Indexed Long [d],y (Read) |                                                                                                    |                                                                                                              |         |
| Cycle                                     | R/W                                                                                                | Desc.                                                                                                        |         |
| -X.2                                      | Read PC:PB                                                                                         | Store as OpCode.                                                                                             |         |
| 1.1                                       |                                                                                                    | Inc. <b>PC</b> .                                                                                             |         |
| 1.2                                       | Read <b>PC:PB</b>                                                                                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                   |         |
| 2.1                                       |                                                                                                    | Inc. <b>PC</b> .                                                                                             |         |
| 2.2                                       |                                                                                                    |                                                                                                              |         |
| 3.1                                       |                                                                                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                            |         |
| 3.2                                       | Read Address                                                                                       | Store as Pointer.                                                                                            |         |
| 4.1                                       |                                                                                                    |                                                                                                              |         |
| 4.2                                       | If <b>D.L</b> is not 0, read (Address + 1), otherwise read (Address & \$FF00)   ui8(Address.L + 1) | Store as Pointer.H.                                                                                          |         |
| 5.1                                       |                                                                                                    |                                                                                                              |         |
| 5.2                                       | If <b>D.L</b> is not 0, read (Address + 2), otherwise read (Address & \$FF00)   ui8(Address.L + 2) | Store as Bank.                                                                                               |         |
| 6.1                                       |                                                                                                    | Perform Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>Bank += (Tmp >> 16).                     |         |
| 6.2                                       | Read Pointer:Bank                                                                                  | Store as Operand.                                                                                            |         |
| 7.1                                       |                                                                                                    | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |         |
| 7.2                                       | Read <b>PC:PB</b>                                                                                  | Store as OpCode.                                                                                             |         |
| +X.1                                      |                                                                                                    |                                                                                                              |         |



|          |                   |                              |
|----------|-------------------|------------------------------|
| CLC (18) | Cycles: 2         | Size: 1                      |
| Implied  |                   |                              |
| Cycle    | R/W               | Desc.                        |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.             |
| 1.1      |                   | Inc. <b>PC</b> .             |
| 1.2      | Read <b>PC:PB</b> | Discard.                     |
| 2.1      |                   | Sets the <b>C</b> flag to 0. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.             |
| +X.1     |                   |                              |

|                    |                     |                                                                                                                                                                                        |
|--------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA (19) Cycles: 4 |                     | Size: 3                                                                                                                                                                                |
| Absolute, Y (Read) |                     |                                                                                                                                                                                        |
| Cycle              | R/W                 | Desc.                                                                                                                                                                                  |
| -X.2               | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                       |
| 1.1                |                     | Inc. <b>PC</b> .                                                                                                                                                                       |
| 1.2                | Read <b>PC:PB</b> . | Store as Pointer.                                                                                                                                                                      |
| 2.1                |                     | Inc. <b>PC</b> .                                                                                                                                                                       |
| 2.2                | Read <b>PC:PB</b> . | Store as Pointer.H.                                                                                                                                                                    |
| 3.1                |                     | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16). |
| 3.2                |                     |                                                                                                                                                                                        |
| 4.1                |                     |                                                                                                                                                                                        |
| 4.2                | Read Pointer:Bank   | Store as Operand.                                                                                                                                                                      |
| 5.1                |                     | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L.                                                                          |
| 5.2                | Read <b>PC:PB</b> . | Store as OpCode.                                                                                                                                                                       |
| +X.1               |                     |                                                                                                                                                                                        |

|          |                   |                                                                                                           |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------|
| INC (1A) | Cycles: 2         | Size: 1                                                                                                   |
| Implied  |                   |                                                                                                           |
| Cycle    | R/W               | Desc.                                                                                                     |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| 1.1      |                   | Inc. <b>PC</b> .                                                                                          |
| 1.2      | Read <b>PC:PB</b> | Discard.                                                                                                  |
| 2.1      |                   | Perform <b>A.L</b> += 1, set <b>N</b> based off ( <b>A.L</b> & \$80), set <b>Z</b> based off <b>A.L</b> . |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| +X.1     |                   |                                                                                                           |

|          |                   |                             |
|----------|-------------------|-----------------------------|
| TSC (1B) | Cycles: 2         | Size: 1                     |
| Implied  |                   |                             |
| Cycle    | R/W               | Desc.                       |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.            |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .            |
| 1.2      |                   | Discard.                    |
| 2.1      |                   | Sets <b>S</b> to <b>A</b> . |
| 2.2      |                   | Store as OpCode.            |
| +X.1     |                   |                             |

|                              |                             |                                                                                    |
|------------------------------|-----------------------------|------------------------------------------------------------------------------------|
| TRB (1C)                     | Cycles: 6                   | Size: 3                                                                            |
| Absolute (Read/Modify/Write) |                             |                                                                                    |
| Cycle                        | R/W                         | Desc.                                                                              |
| -X.2                         | Read <b>PC:PB</b>           | Store as OpCode.                                                                   |
| 1.1                          |                             | Inc. <b>PC</b> .                                                                   |
| 1.2                          | Read <b>PC:PB</b>           | Store as Address.                                                                  |
| 2.1                          |                             | Inc. <b>PC</b> .                                                                   |
| 2.2                          | Read <b>PC:PB</b>           | Store as Address.L.                                                                |
| 3.1                          |                             | Inc. <b>PC</b> .                                                                   |
| 3.2                          | Read Address: <b>DB</b>     | Store as Operand.                                                                  |
| 4.1                          |                             |                                                                                    |
| 4.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                   |
| 5.1                          |                             | Set <b>Z</b> based off ( <b>A</b> .L & Operand.L). Perform Operand &= ~ <b>A</b> . |
| 5.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                   |
| 6.1                          |                             |                                                                                    |
| 6.2                          | Read <b>PC:PB</b>           | Store as OpCode.                                                                   |
| +X.1                         |                             |                                                                                    |

|                    |                     |                                                                                                                                                                                        |
|--------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA (1D) Cycles: 4 |                     | Size: 3                                                                                                                                                                                |
| Absolute, X (Read) |                     |                                                                                                                                                                                        |
| Cycle              | R/W                 | Desc.                                                                                                                                                                                  |
| -X.2               | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                       |
| 1.1                |                     | Inc. <b>PC</b> .                                                                                                                                                                       |
| 1.2                | Read <b>PC:PB</b> . | Store as Pointer.                                                                                                                                                                      |
| 2.1                |                     | Inc. <b>PC</b> .                                                                                                                                                                       |
| 2.2                | Read <b>PC:PB</b> . | Store as Pointer.H.                                                                                                                                                                    |
| 3.1                |                     | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16). |
| 3.2                |                     |                                                                                                                                                                                        |
| 4.1                |                     |                                                                                                                                                                                        |
| 4.2                | Read Pointer:Bank   | Store as Operand.                                                                                                                                                                      |
| 5.1                |                     | Perform <b>A</b>  = Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> .                                                                           |
| 5.2                | Read <b>PC:PB</b> . | Store as OpCode.                                                                                                                                                                       |
| +X.1               |                     |                                                                                                                                                                                        |

|                                |                       |                                                                                                                                                                         |
|--------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASL (1E)                       | Cycles: 7             | Size: 3                                                                                                                                                                 |
| Absolute, X (Read/Modiy/Write) |                       |                                                                                                                                                                         |
| Cycle                          | R/W                   | Desc.                                                                                                                                                                   |
| -X.2                           | Read <b>PC:PB</b>     | Store as OpCode.                                                                                                                                                        |
| 1.1                            |                       | Inc. <b>PC</b> .                                                                                                                                                        |
| 1.2                            | Read <b>PC:PB</b> .   | Store as Pointer.                                                                                                                                                       |
| 2.1                            |                       | Inc. <b>PC</b> .                                                                                                                                                        |
| 2.2                            | Read <b>PC:PB</b> .   | Store as Pointer.H.                                                                                                                                                     |
| 3.1                            |                       | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\text{DB} + (\text{Tmp} \gg 16)$ . |
| 3.2                            |                       |                                                                                                                                                                         |
| 4.1                            |                       |                                                                                                                                                                         |
| 4.2                            | Read Pointer:Bank     | Store as Operand.                                                                                                                                                       |
| 5.1                            |                       |                                                                                                                                                                         |
| 5.2                            | Write to Pointer:Bank | Write Operand.L.                                                                                                                                                        |
| 6.1                            |                       | Set <b>C</b> based off (Operand.L & \$80). Perform $\text{Operand.L} \leq 1$ . Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L.              |
| 6.2                            | Write to Pointer:Bank | Write Operand.L.                                                                                                                                                        |
| 7.1                            |                       |                                                                                                                                                                         |
| 7.2                            | Read <b>PC:PB</b> .   | Store as OpCode.                                                                                                                                                        |
| +X.1                           |                       |                                                                                                                                                                         |

|                         |                     |                                                                                                                            |
|-------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| ORA (1F) Cycles: 5      |                     | Size: 4                                                                                                                    |
| Absolute Long, X (Read) |                     |                                                                                                                            |
| Cycle                   | R/W                 | Desc.                                                                                                                      |
| -X.2                    | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                           |
| 1.1                     |                     | Inc. <b>PC</b> .                                                                                                           |
| 1.2                     | Read <b>PC:PB</b> . | Store as Pointer.                                                                                                          |
| 2.1                     |                     | Inc. <b>PC</b> .                                                                                                           |
| 2.2                     | Read <b>PC:PB</b> . | Store as Pointer.H.                                                                                                        |
| 3.1                     |                     | Inc. <b>PC</b> .                                                                                                           |
| 3.2                     | Read <b>PC:PB</b> . | Stores as Bank.                                                                                                            |
| 4.1                     |                     | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .                                           |
|                         |                     | $\text{Pointer} = \text{ui16}(\text{Tmp})$ .                                                                               |
|                         |                     | $\text{Bank} += (\text{Tmp} \gg 16)$ .                                                                                     |
| 4.2                     | Read Pointer:Bank   | Store as Operand.                                                                                                          |
| 5.1                     |                     | Perform $\text{A} \mid = \text{Operand}$ . Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |
| 5.2                     | Read <b>PC:PB</b> . | Store as OpCode.                                                                                                           |
| +X.1                    |                     |                                                                                                                            |



|          |                         |                                                  |
|----------|-------------------------|--------------------------------------------------|
| JSR (20) | Cycles: 6               | Size: 3                                          |
| Absolute |                         |                                                  |
| Cycle    | R/W                     | Desc.                                            |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.                                 |
| 1.1      |                         | Inc. <b>PC</b> .                                 |
| 1.2      | Read <b>PC:PB</b>       | Store as Address.                                |
| 2.1      |                         | Inc. <b>PC</b> .                                 |
| 2.2      | Read <b>PC:PB</b>       | Store as Address.L.                              |
| 3.1      |                         | Inc. <b>PC</b> .                                 |
| 3.2      | Read <b>PC:PB</b>       | Discard.                                         |
| 4.1      |                         | Dec. <b>PC</b> .                                 |
| 4.2      | Write to (S+0)   \$0100 | Push <b>PC</b> .H onto stack.                    |
| 5.1      |                         |                                                  |
| 5.2      | Write to (S-1)   \$0100 | Push <b>PC</b> .L onto stack.                    |
| 6.1      |                         | Dec. <b>S</b> by 2. Perform <b>PC</b> = Address. |
| 6.2      | Read <b>PC:PB</b>       | Store as OpCode.                                 |
| +X.1     |                         |                                                  |

|                                      |                         |                                                                                                                 |
|--------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| AND (21)                             | Cycles: 6               | Size: 2                                                                                                         |
| Direct Indexed Indirect (d,x) (Read) |                         |                                                                                                                 |
| Cycle                                | R/W                     | Desc.                                                                                                           |
| -X.2                                 | Read <b>PC:PB</b>       | Store as OpCode.                                                                                                |
| 1.1                                  |                         | Inc. <b>PC</b> .                                                                                                |
| 1.2                                  | Read <b>PC:PB</b> .     | Store as Pointer.                                                                                               |
| 2.1                                  |                         | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> . Address.L = Pointer.L. Address.H = <b>D</b> .H. |
| 2.2                                  |                         | If <b>D</b> .L is 0, skip the next cycle.                                                                       |
| 3.1                                  |                         | Address.H = Pointer.H (fixes high byte of address).                                                             |
| 3.2                                  |                         |                                                                                                                 |
| 4.1                                  |                         |                                                                                                                 |
| 4.2                                  | Read Address            | Store as Pointer.                                                                                               |
| 5.1                                  |                         |                                                                                                                 |
| 5.2                                  | Read Address + 1        | Store as Pointer.H.                                                                                             |
| 6.1                                  |                         |                                                                                                                 |
| 6.2                                  | Read Pointer: <b>DB</b> | Store as Operand.                                                                                               |
| 7.1                                  |                         | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L.   |
| 7.2                                  | Read <b>PC:PB</b> .     | Store as OpCode.                                                                                                |
| +X.1                                 |                         |                                                                                                                 |

|               |                                   |                                                                          |
|---------------|-----------------------------------|--------------------------------------------------------------------------|
| JSL (22)      | Cycles: 8                         | Size: 4                                                                  |
| Absolute Long |                                   |                                                                          |
| Cycle         | R/W                               | Desc.                                                                    |
| -X.2          | Read <b>PC:PB</b>                 | Store as OpCode.                                                         |
| 1.1           |                                   | Inc. <b>PC</b> .                                                         |
| 1.2           | Read <b>PC:PB</b>                 | Store as Address.                                                        |
| 2.1           |                                   | Inc. <b>PC</b> .                                                         |
| 2.2           | Read <b>PC:PB</b>                 | Store as Address.L.                                                      |
| 3.1           |                                   | Inc. <b>PC</b> .                                                         |
| 3.2           | Write to ( <b>S</b> +0)   \$0100  | Push <b>PB</b> onto stack.                                               |
| 4.1           |                                   |                                                                          |
| 4.2           | Read ( <b>S</b> +0)   \$0100      | Discard.                                                                 |
| 5.1           |                                   |                                                                          |
| 5.2           | Read <b>PC:PB</b>                 | Stores as Bank.                                                          |
| 6.1           |                                   |                                                                          |
| 6.2           | Write to (( <b>S</b>   \$0100)-1) | Push <b>PC</b> .H onto stack.                                            |
| 7.1           |                                   |                                                                          |
| 7.2           | Write to (( <b>S</b>   \$0100)-2) | Push <b>PC</b> .L onto stack.                                            |
| 8.1           |                                   | Dec. <b>S</b> by 3. Copy Address to <b>PC</b> . Copy Bank to <b>PB</b> . |
| 8.2           | Read <b>PC:PB</b>                 | Store as OpCode.                                                         |
| +X.1          |                                   |                                                                          |

|                       |                     |                                                                                                               |
|-----------------------|---------------------|---------------------------------------------------------------------------------------------------------------|
| AND (23)              | Cycles: 4           | Size: 2                                                                                                       |
| Stack Relative (Read) |                     |                                                                                                               |
| Cycle                 | R/W                 | Desc.                                                                                                         |
| -X.2                  | Read <b>PC:PB</b>   | Store as OpCode.                                                                                              |
| 1.1                   |                     | Inc. <b>PC</b> .                                                                                              |
| 1.2                   | Read <b>PC:PB</b> . | Store as Pointer.                                                                                             |
| 2.1                   |                     | Inc. <b>PC</b> . Set Address to Pointer + ( <b>S</b> .L   \$0100).                                            |
| 2.2                   |                     |                                                                                                               |
| 3.1                   |                     |                                                                                                               |
| 3.2                   | Read Address        | Store as Operand.                                                                                             |
| 4.1                   |                     | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |
| 4.2                   | Read <b>PC:PB</b> . | Store as OpCode.                                                                                              |
| +X.1                  |                     |                                                                                                               |

|               |                   |                                                                                                                                             |
|---------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (24)      | Cycles: 3         | Size: 2                                                                                                                                     |
| Direct (Read) |                   |                                                                                                                                             |
| Cycle         | R/W               | Desc.                                                                                                                                       |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                            |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                            |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                  |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                            |
| 2.2           |                   |                                                                                                                                             |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                           |
| 3.2           | Read Address      | Store as Operand.                                                                                                                           |
| 4.1           |                   | Set <b>Z</b> based off ( <b>A.L</b> & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40). |
| 4.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                            |
| +X.1          |                   |                                                                                                                                             |

|               |                   |                                                                                                              |
|---------------|-------------------|--------------------------------------------------------------------------------------------------------------|
| AND (25)      | Cycles: 3         | Size: 2                                                                                                      |
| Direct (Read) |                   |                                                                                                              |
| Cycle         | R/W               | Desc.                                                                                                        |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                             |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                   |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                             |
| 2.2           |                   |                                                                                                              |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                            |
| 3.2           | Read Address      | Store as Operand.                                                                                            |
| 4.1           |                   | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |
| 4.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |
| +X.1          |                   |                                                                                                              |

|                            |                   |                                                                                                                                                                          |         |
|----------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ROL (26)                   |                   | Cycles: 5                                                                                                                                                                | Size: 2 |
| Direct (Read/Modify/Write) |                   |                                                                                                                                                                          |         |
| Cycle                      | R/W               | Desc.                                                                                                                                                                    |         |
| -X.2                       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                         |         |
| 1.1                        |                   | Inc. <b>PC</b> .                                                                                                                                                         |         |
| 1.2                        | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                               |         |
| 2.1                        |                   | Inc. <b>PC</b> .                                                                                                                                                         |         |
| 2.2                        |                   |                                                                                                                                                                          |         |
| 3.1                        |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                        |         |
| 3.2                        | Read Address      | Store as Operand.                                                                                                                                                        |         |
| 4.1                        |                   |                                                                                                                                                                          |         |
| 4.2                        | Write to Address  | Write Operand.L.                                                                                                                                                         |         |
| 5.1                        |                   | Set <b>C</b> based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1)   <b>C</b> . Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L. |         |
| 5.2                        | Write to Address  | Write Operand.L.                                                                                                                                                         |         |
| 6.1                        |                   |                                                                                                                                                                          |         |
| 6.2                        | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                         |         |
| +X.1                       |                   |                                                                                                                                                                          |         |

|                             |                   |                                                                                                              |         |
|-----------------------------|-------------------|--------------------------------------------------------------------------------------------------------------|---------|
| AND (27)                    |                   | Cycles: 6                                                                                                    | Size: 2 |
| Direct Indirect Long (Read) |                   |                                                                                                              |         |
| Cycle                       | R/W               | Desc.                                                                                                        |         |
| -X.2                        | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |         |
| 1.1                         |                   | Inc. <b>PC</b> .                                                                                             |         |
| 1.2                         | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                   |         |
| 2.1                         |                   | Inc. <b>PC</b> .                                                                                             |         |
| 2.2                         |                   |                                                                                                              |         |
| 3.1                         |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                            |         |
| 3.2                         | Read Address      | Store as Pointer.                                                                                            |         |
| 4.1                         |                   |                                                                                                              |         |
| 4.2                         | Read Address + 1  | Store as Pointer.H.                                                                                          |         |
| 5.1                         |                   |                                                                                                              |         |
| 5.2                         | Read Address + 2  | Store as Bank.                                                                                               |         |
| 6.1                         |                   |                                                                                                              |         |
| 6.2                         | Read Pointer:Bank | Store as Operand.                                                                                            |         |
| 7.1                         |                   | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |         |
| 7.2                         | Read <b>PC:PB</b> | Store as OpCode.                                                                                             |         |
| +X.1                        |                   |                                                                                                              |         |



|          |                        |                                                                                            |
|----------|------------------------|--------------------------------------------------------------------------------------------|
| PLP (28) | Cycles: 4              | Size: 1                                                                                    |
| Implied  |                        |                                                                                            |
| Cycle    | R/W                    | Desc.                                                                                      |
| -X.2     | Read <b>PC:PB</b>      | Store as OpCode.                                                                           |
| 1.1      | Read <b>S</b>   \$0100 | Inc. <b>PC</b> .                                                                           |
| 1.2      |                        |                                                                                            |
| 2.1      |                        |                                                                                            |
| 2.2      |                        |                                                                                            |
| 3.1      |                        | Inc. <b>S</b> by 1.                                                                        |
| 3.2      |                        | Store as Operand.L.                                                                        |
| 4.1      |                        | <b>P</b> = Operand.L, set <b>X</b> and <b>M</b> flags, set <b>X.H</b> and <b>Y.H</b> to 0. |
| 4.2      | Read <b>PC:PB</b>      | Store as OpCode.                                                                           |
| +X.1     |                        |                                                                                            |

|           |                   |                                                                                                                               |
|-----------|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| AND (29)  | Cycles: 2         | Size: 2                                                                                                                       |
| Immediate |                   |                                                                                                                               |
| Cycle     | R/W               | Desc.                                                                                                                         |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                              |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                              |
| 1.2       | Read <b>PC:PB</b> | Store as Operand.                                                                                                             |
| 2.1       |                   | Inc. <b>PC</b> . Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |
| 2.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                              |
| +X.1      |                   |                                                                                                                               |

|          |                   |                                                                                                                                                                                   |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (2A) | Cycles: 2         | Size: 1                                                                                                                                                                           |
| Implied  |                   |                                                                                                                                                                                   |
| Cycle    | R/W               | Desc.                                                                                                                                                                             |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                  |
| 1.1      |                   | Inc. <b>PC</b> .                                                                                                                                                                  |
| 1.2      | Read <b>PC:PB</b> | Discard.                                                                                                                                                                          |
| 2.1      |                   | Set <b>C</b> based off ( <b>A.L</b> & \$80). Perform <b>A.L</b> = ( <b>A.L</b> << 1)   <b>C</b> . Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                  |
| +X.1     |                   |                                                                                                                                                                                   |

|          |                               |                                                                                                                              |
|----------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| PLD (2B) | Cycles: 5                     | Size: 1                                                                                                                      |
| Implied  |                               |                                                                                                                              |
| Cycle    | R/W                           | Desc.                                                                                                                        |
| -X.2     | Read <b>PC:PB</b>             | Store as OpCode.                                                                                                             |
| 1.1      | Read (( <b>S</b>   \$0100)+1) | Inc. <b>PC</b> .                                                                                                             |
| 1.2      |                               |                                                                                                                              |
| 2.1      |                               |                                                                                                                              |
| 2.2      |                               |                                                                                                                              |
| 3.1      |                               |                                                                                                                              |
| 3.2      |                               | Store as Operand.L.                                                                                                          |
| 4.1      |                               |                                                                                                                              |
| 4.2      |                               | Store as Operand.H.                                                                                                          |
| 5.1      |                               | Inc. <b>S</b> by 2. Set <b>D</b> to Operand. Set <b>N</b> based off ( <b>D</b> .H & \$80). Set <b>Z</b> based off <b>D</b> . |
| 5.2      |                               | Store as OpCode.                                                                                                             |
| +X.1     |                               |                                                                                                                              |

|                 |                         |                                                                                                                                              |
|-----------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (2C)        | Cycles: 4               | Size: 3                                                                                                                                      |
| Absolute (Read) |                         |                                                                                                                                              |
| Cycle           | R/W                     | Desc.                                                                                                                                        |
| -X.2            | Read <b>PC:PB</b>       | Store as OpCode.                                                                                                                             |
| 1.1             |                         | Inc. <b>PC</b> .                                                                                                                             |
| 1.2             | Read <b>PC:PB</b>       | Store as Address.                                                                                                                            |
| 2.1             |                         | Inc. <b>PC</b> .                                                                                                                             |
| 2.2             | Read <b>PC:PB</b>       | Store as Address.L.                                                                                                                          |
| 3.1             |                         | Inc. <b>PC</b> .                                                                                                                             |
| 3.2             | Read Address: <b>DB</b> | Store as Operand.                                                                                                                            |
| 4.1             |                         | Set <b>Z</b> based off ( <b>A</b> .L & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40). |
| 4.2             | Read <b>PC:PB</b>       | Store as OpCode.                                                                                                                             |
| +X.1            |                         |                                                                                                                                              |

|                 |                         |                                                                                                                                              |
|-----------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (2C)        | Cycles: 4               | Size: 3                                                                                                                                      |
| Absolute (Read) |                         |                                                                                                                                              |
| Cycle           | R/W                     | Desc.                                                                                                                                        |
| -X.2            | Read <b>PC:PB</b>       | Store as OpCode.                                                                                                                             |
| 1.1             |                         | Inc. <b>PC</b> .                                                                                                                             |
| 1.2             | Read <b>PC:PB</b>       | Store as Address.                                                                                                                            |
| 2.1             |                         | Inc. <b>PC</b> .                                                                                                                             |
| 2.2             | Read <b>PC:PB</b>       | Store as Address.L.                                                                                                                          |
| 3.1             |                         | Inc. <b>PC</b> .                                                                                                                             |
| 3.2             | Read Address: <b>DB</b> | Store as Operand.                                                                                                                            |
| 4.1             |                         | Set <b>Z</b> based off ( <b>A</b> .L & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40). |
| 4.2             | Read <b>PC:PB</b>       | Store as OpCode.                                                                                                                             |
| +X.1            |                         |                                                                                                                                              |

|                 |                         |                                                                                                               |
|-----------------|-------------------------|---------------------------------------------------------------------------------------------------------------|
| AND (2D)        | Cycles: 4               | Size: 3                                                                                                       |
| Absolute (Read) |                         |                                                                                                               |
| Cycle           | R/W                     | Desc.                                                                                                         |
| -X.2            | Read <b>PC:PB</b>       | Store as OpCode.                                                                                              |
| 1.1             |                         | Inc. <b>PC</b> .                                                                                              |
| 1.2             | Read <b>PC:PB</b>       | Store as Address.                                                                                             |
| 2.1             |                         | Inc. <b>PC</b> .                                                                                              |
| 2.2             | Read <b>PC:PB</b>       | Store as Address.L.                                                                                           |
| 3.1             |                         | Inc. <b>PC</b> .                                                                                              |
| 3.2             | Read Address: <b>DB</b> | Store as Operand.                                                                                             |
| 4.1             |                         | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |
| 4.2             | Read <b>PC:PB</b>       | Store as OpCode.                                                                                              |
| +X.1            |                         |                                                                                                               |

|                              |                             |                                                                                                                                                                          |
|------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (2E)                     | Cycles: 6                   | Size: 3                                                                                                                                                                  |
| Absolute (Read/Modify/Write) |                             |                                                                                                                                                                          |
| Cycle                        | R/W                         | Desc.                                                                                                                                                                    |
| -X.2                         | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                         |
| 1.1                          |                             | Inc. <b>PC</b> .                                                                                                                                                         |
| 1.2                          | Read <b>PC:PB</b>           | Store as Address.                                                                                                                                                        |
| 2.1                          |                             | Inc. <b>PC</b> .                                                                                                                                                         |
| 2.2                          | Read <b>PC:PB</b>           | Store as Address.L.                                                                                                                                                      |
| 3.1                          |                             | Inc. <b>PC</b> .                                                                                                                                                         |
| 3.2                          | Read Address: <b>DB</b>     | Store as Operand.                                                                                                                                                        |
| 4.1                          |                             |                                                                                                                                                                          |
| 4.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                                                                                                         |
| 5.1                          |                             | Set <b>C</b> based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1)   <b>C</b> . Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L. |
| 5.2                          | Write to Address: <b>DB</b> | Write Operand.L.                                                                                                                                                         |
| 6.1                          |                             |                                                                                                                                                                          |
| 6.2                          | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                         |
| +X.1                         |                             |                                                                                                                                                                          |



|                      |                   |                                                                                                               |         |
|----------------------|-------------------|---------------------------------------------------------------------------------------------------------------|---------|
| AND (2F)             |                   | Cycles: 5                                                                                                     | Size: 4 |
| Absolute Long (Read) |                   |                                                                                                               |         |
| Cycle                | R/W               | Desc.                                                                                                         |         |
| -X.2                 | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| 1.1                  |                   | Inc. <b>PC</b> .                                                                                              |         |
| 1.2                  | Read <b>PC:PB</b> | Store as Address.                                                                                             |         |
| 2.1                  |                   | <b>Inc. PC</b> .                                                                                              |         |
| 2.2                  | Read <b>PC:PB</b> | Store as Address.L.                                                                                           |         |
| 3.1                  |                   | <b>Inc. PC</b> .                                                                                              |         |
| 3.2                  | Read <b>PC:PB</b> | Stores as Bank.                                                                                               |         |
| 4.1                  |                   | <b>Inc. PC</b> .                                                                                              |         |
| 4.2                  | Read Address:Bank | Store as Operand.                                                                                             |         |
| 5.1                  |                   | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |         |
| 5.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| +X.1                 |                   |                                                                                                               |         |

|          |                   |                                                                                                                                                                     |
|----------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BMI (30) | Cycles: 2         | Size: 2                                                                                                                                                             |
| Relative |                   |                                                                                                                                                                     |
| Cycle    | R/W               | Desc.                                                                                                                                                               |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                    |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> . Decide to branch if <b>N</b> is 1.                                                                                                                 |
| 1.2      |                   | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> , BoundaryCrossed = Address.H != <b>PC</b> .H. If not branching or not BoundaryCrossed, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 4.2).                                                                                                    |
| 2.2      |                   |                                                                                                                                                                     |
| 3.1      |                   | Set <b>PC</b> .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).                                                                                |
| 3.2      |                   | Poll interrupts.                                                                                                                                                    |
| 4.1      |                   | Set <b>PC</b> .H to Address.H                                                                                                                                       |
| 4.2      |                   | Store as OpCode.                                                                                                                                                    |
| +X.1     |                   |                                                                                                                                                                     |

|                                      |                   |                                                                                                                                                                       |         |
|--------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (31)                             |                   | Cycles: 5                                                                                                                                                             | Size: 2 |
| Direct Indirect Indexed (d),y (Read) |                   |                                                                                                                                                                       |         |
| Cycle                                | R/W               | Desc.                                                                                                                                                                 |         |
| -X.2                                 | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                      |         |
| 1.1                                  |                   | Inc. <b>PC</b> .                                                                                                                                                      |         |
| 1.2                                  | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                            |         |
| 2.1                                  |                   | Inc. <b>PC</b> .                                                                                                                                                      |         |
| 2.2                                  |                   |                                                                                                                                                                       |         |
| 3.1                                  |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                     |         |
| 3.2                                  | Read Address      | Store as Pointer.                                                                                                                                                     |         |
| 4.1                                  |                   |                                                                                                                                                                       |         |
| 4.2                                  | Read Address + 1  | Store as Pointer.H.                                                                                                                                                   |         |
| 5.1                                  |                   | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16). |         |
| 5.2                                  |                   |                                                                                                                                                                       |         |
| 6.1                                  |                   |                                                                                                                                                                       |         |
| 6.2                                  | Read Pointer:Bank | Store as Operand.                                                                                                                                                     |         |
| 7.1                                  |                   | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> .                                                          |         |
| 7.2                                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                      |         |
| +X.1                                 |                   |                                                                                                                                                                       |         |

|                        |                         |                                                                                                              |
|------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------|
| AND (32)               | Cycles: 5               | Size: 2                                                                                                      |
| Direct Indirect (Read) |                         |                                                                                                              |
| Cycle                  | R/W                     | Desc.                                                                                                        |
| -X.2                   | Read <b>PC:PB</b>       | Store as OpCode.                                                                                             |
| 1.1                    |                         | Inc. <b>PC</b> .                                                                                             |
| 1.2                    | Read <b>PC:PB</b>       | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                   |
| 2.1                    |                         | Inc. <b>PC</b> .                                                                                             |
| 2.2                    |                         |                                                                                                              |
| 3.1                    |                         | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                            |
| 3.2                    | Read Address            | Store as Pointer.                                                                                            |
| 4.1                    |                         |                                                                                                              |
| 4.2                    | Read Address + 1        | Store as Pointer.H.                                                                                          |
| 5.1                    |                         |                                                                                                              |
| 5.2                    | Read Pointer: <b>DB</b> | Store as Operand.                                                                                            |
| 6.1                    |                         | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |
| 6.2                    | Read <b>PC:PB</b>       | Store as OpCode.                                                                                             |
| +X.1                   |                         |                                                                                                              |

|                                        |                     |                                                                                                               |         |
|----------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------|---------|
| AND (33)                               |                     | Cycles: 7                                                                                                     | Size: 2 |
| Stack Relative Indirect Indexed (Read) |                     |                                                                                                               |         |
| Cycle                                  | R/W                 | Desc.                                                                                                         |         |
| -X.2                                   | Read <b>PC:PB</b>   | Store as OpCode.                                                                                              |         |
| 1.1                                    | Read <b>PC:PB</b> . | Inc. <b>PC</b> .                                                                                              |         |
| 1.2                                    |                     | Store as Pointer.                                                                                             |         |
| 2.1                                    |                     | Inc. <b>PC</b> . Set Address to Pointer + ( <b>S</b> .L   \$0100).                                            |         |
| 2.2                                    |                     |                                                                                                               |         |
| 3.1                                    |                     |                                                                                                               |         |
| 3.2                                    |                     | Store as Pointer.                                                                                             |         |
| 4.1                                    |                     |                                                                                                               |         |
| 4.2                                    |                     | Store as Pointer.H.                                                                                           |         |
| 5.1                                    |                     | Perform Tmp = ui32(Pointer + <b>Y</b> ).                                                                      |         |
| 5.2                                    |                     | Pointer = ui16(Tmp).                                                                                          |         |
| 6.1                                    |                     | Bank = <b>DB</b> + (Tmp >> 16).                                                                               |         |
| 6.2                                    | Read Pointer:Bank   | Store as Operand.                                                                                             |         |
| 7.1                                    | Read <b>PC:PB</b> . | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |         |
| 7.2                                    |                     | Store as OpCode.                                                                                              |         |
| +X.1                                   |                     |                                                                                                               |         |

|                  |                     |                                                                                                                                                                                          |
|------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (34)         | Cycles: 4           | Size: 2                                                                                                                                                                                  |
| Direct, X (Read) |                     |                                                                                                                                                                                          |
| Cycle            | R/W                 | Desc.                                                                                                                                                                                    |
| -X.2             | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                         |
| 1.1              | Read <b>PC:PB</b> . | Inc. <b>PC</b> .                                                                                                                                                                         |
| 1.2              |                     | Store as Operand.                                                                                                                                                                        |
| 2.1              |                     | Inc. <b>PC</b> .                                                                                                                                                                         |
| 2.2              |                     |                                                                                                                                                                                          |
| 3.1              |                     | If <b>D.L</b> is 0, Address.L = Operand + <b>X.L</b> + <b>D.L</b> , Address.H = <b>D.H</b> , otherwise Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2              |                     |                                                                                                                                                                                          |
| 4.1              |                     |                                                                                                                                                                                          |
| 4.2              | Read Address        | Store as Operand.                                                                                                                                                                        |
| 5.1              | Read <b>PC:PB</b> . | Set <b>Z</b> based off ( <b>A.L</b> & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40).                                              |
| 5.2              |                     | Store as OpCode.                                                                                                                                                                         |
| +X.1             |                     |                                                                                                                                                                                          |

|                  |                   |                                                                                                                                                                                          |         |
|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (35)         |                   | Cycles: 4                                                                                                                                                                                | Size: 2 |
| Direct, X (Read) |                   |                                                                                                                                                                                          |         |
| Cycle            | R/W               | Desc.                                                                                                                                                                                    |         |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                         |         |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                         |         |
| 1.2              |                   | Store as Operand.                                                                                                                                                                        |         |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                         |         |
| 2.2              |                   |                                                                                                                                                                                          |         |
| 3.1              |                   | If <b>D.L</b> is 0, Address.L = Operand + <b>X.L</b> + <b>D.L</b> , Address.H = <b>D.H</b> , otherwise Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |         |
| 3.2              |                   |                                                                                                                                                                                          |         |
| 4.1              |                   |                                                                                                                                                                                          |         |
| 4.2              | Read Address      | Store as Operand.                                                                                                                                                                        |         |
| 5.1              | Read <b>PC:PB</b> | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> .                                                                             |         |
| 5.2              |                   | Store as OpCode.                                                                                                                                                                         |         |
| +X.1             |                   |                                                                                                                                                                                          |         |

| ROL (36)                                          | Cycles: 6         | Size: 2                                                                                                                                                                                  |
|---------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Read/Modify/Write) |                   |                                                                                                                                                                                          |
| Cycle                                             | R/W               | Desc.                                                                                                                                                                                    |
| -X.2                                              | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                         |
| 1.1                                               | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                         |
| 1.2                                               |                   | Store as Operand.                                                                                                                                                                        |
| 2.1                                               |                   | Inc. <b>PC</b> .                                                                                                                                                                         |
| 2.2                                               |                   |                                                                                                                                                                                          |
| 3.1                                               |                   | If <b>D.L</b> is 0, Address.L = Operand + <b>X.L</b> + <b>D.L</b> , Address.H = <b>D.H</b> , otherwise Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2                                               |                   |                                                                                                                                                                                          |
| 4.1                                               |                   |                                                                                                                                                                                          |
| 4.2                                               | Read Address      | Store as Operand.                                                                                                                                                                        |
| 5.1                                               | Write to Address  | Write Operand.L.                                                                                                                                                                         |
| 5.2                                               |                   | Set <b>C</b> based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1)   <b>C</b> . Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L.                 |
| 6.1                                               |                   |                                                                                                                                                                                          |
| 6.2                                               | Write to Address  | Write Operand.L.                                                                                                                                                                         |
| 7.1                                               | Read <b>PC:PB</b> |                                                                                                                                                                                          |
| 7.2                                               |                   | Store as OpCode.                                                                                                                                                                         |
| +X.1                                              |                   |                                                                                                                                                                                          |



| AND (37)                                  |                                                                                                    | Cycles: 6                                                                                                    | Size: 2 |
|-------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------|
| Direct Indirect Indexed Long [d],y (Read) |                                                                                                    |                                                                                                              |         |
| Cycle                                     | R/W                                                                                                | Desc.                                                                                                        |         |
| -X.2                                      | Read <b>PC:PB</b>                                                                                  | Store as OpCode.                                                                                             |         |
| 1.1                                       |                                                                                                    | Inc. <b>PC</b> .                                                                                             |         |
| 1.2                                       | Read <b>PC:PB</b>                                                                                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                   |         |
| 2.1                                       |                                                                                                    | Inc. <b>PC</b> .                                                                                             |         |
| 2.2                                       |                                                                                                    |                                                                                                              |         |
| 3.1                                       |                                                                                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                            |         |
| 3.2                                       | Read Address                                                                                       | Store as Pointer.                                                                                            |         |
| 4.1                                       |                                                                                                    |                                                                                                              |         |
| 4.2                                       | If <b>D.L</b> is not 0, read (Address + 1), otherwise read (Address & \$FF00)   ui8(Address.L + 1) | Store as Pointer.H.                                                                                          |         |
| 5.1                                       |                                                                                                    |                                                                                                              |         |
| 5.2                                       | If <b>D.L</b> is not 0, read (Address + 2), otherwise read (Address & \$FF00)   ui8(Address.L + 2) | Store as Bank.                                                                                               |         |
| 6.1                                       |                                                                                                    | Perform Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>Bank += (Tmp >> 16).                     |         |
| 6.2                                       | Read Pointer:Bank                                                                                  | Store as Operand.                                                                                            |         |
| 7.1                                       |                                                                                                    | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |         |
| 7.2                                       | Read <b>PC:PB</b>                                                                                  | Store as OpCode.                                                                                             |         |
| +X.1                                      |                                                                                                    |                                                                                                              |         |

|          |                   |                              |
|----------|-------------------|------------------------------|
| SEC (38) | Cycles: 2         | Size: 1                      |
| Implied  |                   |                              |
| Cycle    | R/W               | Desc.                        |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.             |
| 1.1      |                   | Inc. <b>PC</b> .             |
| 1.2      | Read <b>PC:PB</b> | Discard.                     |
| 2.1      |                   | Sets the <b>C</b> flag to 1. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.             |
| +X.1     |                   |                              |

|                    |                   |                                                                                                               |         |
|--------------------|-------------------|---------------------------------------------------------------------------------------------------------------|---------|
| AND (39)           |                   | Cycles: 4                                                                                                     | Size: 3 |
| Absolute, Y (Read) |                   |                                                                                                               |         |
| Cycle              | R/W               | Desc.                                                                                                         |         |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| 1.1                |                   | Inc. <b>PC</b> .                                                                                              |         |
| 1.2                | Read <b>PC:PB</b> | Store as Pointer.                                                                                             |         |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                              |         |
| 2.2                | Read <b>PC:PB</b> | Store as Pointer.H.                                                                                           |         |
| 3.1                |                   | Inc. <b>PC</b> . Perform Orig = Pointer.                                                                      |         |
|                    |                   | Tmp = ui32(Pointer + <b>Y</b> ).                                                                              |         |
|                    |                   | Pointer = ui16(Tmp).                                                                                          |         |
|                    |                   | If Orig.H == Pointer.H, skip 2 half-cycles.                                                                   |         |
|                    |                   | Bank = <b>DB</b> + (Tmp >> 16).                                                                               |         |
| 3.2                |                   |                                                                                                               |         |
| 4.1                |                   |                                                                                                               |         |
| 4.2                | Read Pointer:Bank | Store as Operand.                                                                                             |         |
| 5.1                |                   | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |         |
| 5.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| +X.1               |                   |                                                                                                               |         |

|          |                   |                                                                                                           |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------|
| DEC (3A) | Cycles: 2         | Size: 1                                                                                                   |
| Implied  |                   |                                                                                                           |
| Cycle    | R/W               | Desc.                                                                                                     |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| 1.1      |                   | Inc. <b>PC</b> .                                                                                          |
| 1.2      | Read <b>PC:PB</b> | Discard.                                                                                                  |
| 2.1      |                   | Perform <b>A.L</b> -= 1, set <b>N</b> based off ( <b>A.L</b> & \$80), set <b>Z</b> based off <b>A.L</b> . |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| +X.1     |                   |                                                                                                           |

|          |                   |                                                                                                                      |
|----------|-------------------|----------------------------------------------------------------------------------------------------------------------|
| TSC (3B) | Cycles: 2         | Size: 1                                                                                                              |
| Implied  |                   |                                                                                                                      |
| Cycle    | R/W               | Desc.                                                                                                                |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                     |
| 1.1      |                   | Inc. <b>PC</b> .                                                                                                     |
| 1.2      | Read <b>PC:PB</b> | Discard.                                                                                                             |
| 2.1      |                   | Set <b>A</b> to ( <b>S</b>   \$0100). Set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                     |
| +X.1     |                   |                                                                                                                      |

|                    |                   |                                                                                                                                                                                        |
|--------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (3C)           | Cycles: 4         | Size: 3                                                                                                                                                                                |
| Absolute, X (Read) |                   |                                                                                                                                                                                        |
| Cycle              | R/W               | Desc.                                                                                                                                                                                  |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                       |
| 1.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                       |
| 1.2                | Read <b>PC:PB</b> | Store as Pointer.                                                                                                                                                                      |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                       |
| 2.2                | Read <b>PC:PB</b> | Store as Pointer.H.                                                                                                                                                                    |
| 3.1                |                   | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16). |
| 3.2                |                   |                                                                                                                                                                                        |
| 4.1                |                   |                                                                                                                                                                                        |
| 4.2                | Read Pointer:Bank | Store as Operand.                                                                                                                                                                      |
| 5.1                |                   | Set <b>Z</b> based off ( <b>A</b> .L & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40).                                           |
| 5.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                       |
| +X.1               |                   |                                                                                                                                                                                        |

|                    |                   |                                                                                                               |         |
|--------------------|-------------------|---------------------------------------------------------------------------------------------------------------|---------|
| AND (3D)           |                   | Cycles: 4                                                                                                     | Size: 3 |
| Absolute, X (Read) |                   |                                                                                                               |         |
| Cycle              | R/W               | Desc.                                                                                                         |         |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| 1.1                |                   | Inc. <b>PC</b> .                                                                                              |         |
| 1.2                | Read <b>PC:PB</b> | Store as Pointer.                                                                                             |         |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                              |         |
| 2.2                | Read <b>PC:PB</b> | Store as Pointer.H.                                                                                           |         |
| 3.1                |                   | Inc. <b>PC</b> . Perform Orig = Pointer.                                                                      |         |
|                    |                   | Tmp = ui32(Pointer + <b>X</b> ).                                                                              |         |
|                    |                   | Pointer = ui16(Tmp).                                                                                          |         |
|                    |                   | If Orig.H == Pointer.H, skip 2 half-cycles.                                                                   |         |
|                    |                   | Bank = <b>DB</b> + (Tmp >> 16).                                                                               |         |
| 3.2                |                   |                                                                                                               |         |
| 4.1                |                   |                                                                                                               |         |
| 4.2                | Read Pointer:Bank | Store as Operand.                                                                                             |         |
| 5.1                |                   | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L. |         |
| 5.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                              |         |
| +X.1               |                   |                                                                                                               |         |

|                                |                       |                                                                                                                                                                                              |
|--------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (3E)                       | Cycles: 7             | Size: 3                                                                                                                                                                                      |
| Absolute, X (Read/Modiy/Write) |                       |                                                                                                                                                                                              |
| Cycle                          | R/W                   | Desc.                                                                                                                                                                                        |
| -X.2                           | Read <b>PC:PB</b>     | Store as OpCode.                                                                                                                                                                             |
| 1.1                            |                       | Inc. <b>PC</b> .                                                                                                                                                                             |
| 1.2                            | Read <b>PC:PB</b> .   | Store as Pointer.                                                                                                                                                                            |
| 2.1                            |                       | Inc. <b>PC</b> .                                                                                                                                                                             |
| 2.2                            | Read <b>PC:PB</b> .   | Store as Pointer.H.                                                                                                                                                                          |
| 3.1                            |                       | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\text{DB} + (\text{Tmp} \gg 16)$ .                      |
| 3.2                            |                       |                                                                                                                                                                                              |
| 4.1                            |                       |                                                                                                                                                                                              |
| 4.2                            | Read Pointer:Bank     | Store as Operand.                                                                                                                                                                            |
| 5.1                            |                       |                                                                                                                                                                                              |
| 5.2                            | Write to Pointer:Bank | Write Operand.L.                                                                                                                                                                             |
| 6.1                            |                       | Set <b>C</b> based off (Operand.L & \$80). Perform $\text{Operand.L} = (\text{Operand.L} \ll 1) \mid \text{C}$ . Set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L. |
| 6.2                            | Write to Pointer:Bank | Write Operand.L.                                                                                                                                                                             |
| 7.1                            |                       |                                                                                                                                                                                              |
| 7.2                            | Read <b>PC:PB</b> .   | Store as OpCode.                                                                                                                                                                             |
| +X.1                           |                       |                                                                                                                                                                                              |



|                         |                     |                                                                                                              |         |
|-------------------------|---------------------|--------------------------------------------------------------------------------------------------------------|---------|
| AND (3F)                |                     | Cycles: 5                                                                                                    | Size: 4 |
| Absolute Long, X (Read) |                     |                                                                                                              |         |
| Cycle                   | R/W                 | Desc.                                                                                                        |         |
| -X.2                    | Read <b>PC:PB</b>   | Store as OpCode.                                                                                             |         |
| 1.1                     |                     | Inc. <b>PC</b> .                                                                                             |         |
| 1.2                     | Read <b>PC:PB</b> . | Store as Pointer.                                                                                            |         |
| 2.1                     |                     | Inc. <b>PC</b> .                                                                                             |         |
| 2.2                     | Read <b>PC:PB</b> . | Store as Pointer.H.                                                                                          |         |
| 3.1                     |                     | Inc. <b>PC</b> .                                                                                             |         |
| 3.2                     | Read <b>PC:PB</b> . | Stores as Bank.                                                                                              |         |
| 4.1                     |                     | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .                             |         |
|                         |                     | Pointer = $\text{ui16}(\text{Tmp})$ .                                                                        |         |
|                         |                     | Bank += (Tmp >> 16).                                                                                         |         |
| 4.2                     | Read Pointer:Bank   | Store as Operand.                                                                                            |         |
| 5.1                     |                     | Perform <b>A</b> &= Operand. Set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> . |         |
| 5.2                     | Read <b>PC:PB</b> . | Store as OpCode.                                                                                             |         |
| +X.1                    |                     |                                                                                                              |         |